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X-ray Diffraction Analysis

Powder Samples and Epitaxial ZnO Thin Films

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Abstract

X-ray diffraction was used to investigate polycrystalline ZnO and Cu samples, epitaxial ZnO thin films on sapphire substrates, and an unknown oxide powder. The polycrystalline samples were analysed by comparing the measured diffraction patterns with PC-PDF reference data and by determining lattice constants from the assigned reflections. For ZnO, Nelson–Riley extrapolation gave $a = (3.2480 \pm 0.0008)$ Å and $c = (5.2022 \pm 0.0010)$ Å, while the copper sample gave $a = (3.6093 \pm 0.0029)$ Å. Both results agreed well with the corresponding reference values. The ZnO thin films showed strong preferred orientation rather than random polycrystallinity. The film on a-plane sapphire was assigned a [0001] out-of-plane orientation, while the film on r-plane sapphire was assigned a $[11\bar{2}0]$ out-of-plane orientation. Local Gaussian fits and ω-scan rocking curves showed broader ZnO film peaks than sapphire substrate peaks, indicating larger mosaic spread in the films, especially for the r-plane ZnO film. The obtained ZnO film lattice parameters were $c = (5.2037 \pm 0.0001)$ Å and $a = (3.2492 \pm 0.0002)$ Å. The unknown oxide powder was identified as hexagonal Lanthanum Molybdenum Oxide, $La_6Mo_2O_{14}$, after applying a systematic angular correction of approximately $-0.47°$.

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1. Introduction

X-ray diffraction was used as a structural analysis method to investigate crystalline materials with different structural forms and degrees of orientational order. The experiment included polycrystalline bulk samples, epitaxial zinc oxide thin films grown on sapphire substrates, and an unknown oxide powder. The measured diffraction patterns were used to identify crystalline phases, assign diffraction peaks to lattice planes, determine lattice parameters, and evaluate the structural quality of the investigated samples.

For the polycrystalline bulk samples, the diffraction patterns contained several reflections because many differently oriented crystallites contributed to the measurement. The observed peak positions were compared with reference data from the PC-PDF database, allowing the crystal structures and Miller indices of the measured reflections to be assigned. The lattice constants were then determined from the assigned peak positions and corrected for systematic angular deviations by extrapolation methods.

The ZnO thin films on sapphire were studied in order to investigate how epitaxial or strongly textured growth influenced the observed diffraction pattern. In contrast to the polycrystalline reference sample, the thin-film measurements showed only selected ZnO reflections, since only lattice planes parallel to the sample surface were strongly visible in the symmetric diffraction geometry. This made it possible to determine the out-of-plane crystallographic orientation of the films. Additional rocking-curve measurements were performed on selected ZnO and sapphire reflections to compare the peak widths of the film and substrate. These measurements provided information about the crystalline alignment and mosaic spread of the thin films.

Finally, the unknown oxide powder was analysed by comparing its measured diffraction pattern with database reference patterns. The phase identification was based mainly on the agreement between measured and reference peak positions, while relative peak intensities and crystallographic consistency were used as supporting criteria. Overall, the experiment demonstrated how X-ray diffraction could be applied to phase identification, lattice-constant determination, thin-film orientation analysis, and the assessment of crystalline quality.

2. Theoretical Background

2.1 Generation of X-rays

X-rays are electromagnetic waves with wavelengths on the order of interatomic distances in crystalline solids. This makes them suitable for investigating crystal structures, since their wavelengths are comparable to the spacing between lattice planes.

In an X-ray tube, electrons are emitted from a heated cathode and accelerated towards a metal anode by a high voltage. When the electrons hit the anode material, they are rapidly decelerated. Part of their kinetic energy is converted into X-ray radiation. This process produces a continuous background spectrum known as Bremsstrahlung. In addition to this continuous radiation, characteristic X-ray lines are produced when incident electrons remove inner-shell electrons from atoms in the anode. Electrons from higher shells then fill the vacancies, and the energy difference is emitted as X-ray photons.

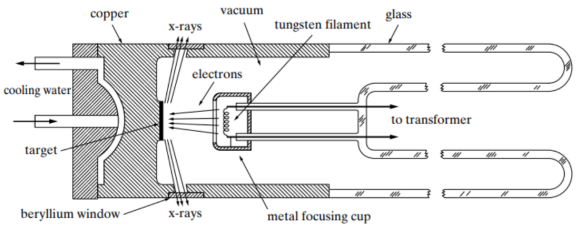


Figure 1: Schematic diagram of an X-ray tube. Electrons emitted from the heated cathode are accelerated towards the metal target, where Bremsstrahlung and characteristic X-ray radiation are generated. Ref. [1].

The X-ray source used in this experiment was a copper anode. Therefore, the dominant characteristic radiation was copper K_α radiation. The finite spectral width and the presence of additional characteristic lines had to be considered during peak assignment, especially when weak extra peaks appeared near strong diffraction maxima.

2.2 Characteristic X-ray Spectrum

The characteristic X-ray spectrum consists of discrete emission lines whose energies depend on the anode material. For a copper anode, the most important lines are the K_α and K_β lines. The K_α line occurs when an electron from the L shell fills a vacancy in the K shell, while the K_β line corresponds to a transition from a higher shell to the K shell.

The copper K_α line is split into two closely spaced components, $K_{\alpha1}$ and $K_{\alpha2}$. This doublet structure can lead to asymmetric diffraction peaks or visible shoulders, especially at larger diffraction angles. In peak analysis, the $K_{\alpha1}$ position is usually used as the main reference position because it is the stronger component of the doublet.

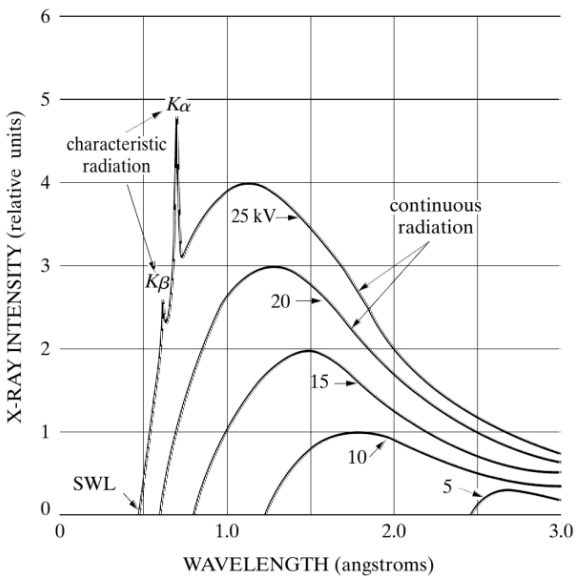


Figure 2: Schematic X-ray emission spectrum showing the continuous Bremsstrahlung background and sharp characteristic lines. The characteristic lines arise from electronic transitions between inner atomic shells. Ref. [1].

Weak additional spectral lines can also appear in the measured diffraction pattern. In this experiment, possible copper K_β and tungsten L_α contributions were considered during peak assignment. These lines do not correspond to additional crystal phases. Instead, they arise from the spectral composition of the X-ray beam and can produce weak diffraction peaks at positions different from the main copper K_α reflections.

2.3 Absorption and Filtering of X-rays

When X-rays pass through matter, part of the radiation is absorbed. The transmitted intensity decreases with increasing absorber thickness according to

$$I = I_0 \exp(-\mu x), \tag{1}$$

where I_0 is the incident intensity, I is the transmitted intensity, μ is the linear absorption coefficient, and x is the absorber thickness. The absorption coefficient depends on the material and on the photon energy.

Absorption effects are important in X-ray diffraction because they influence the measured peak intensities and the penetration depth of the X-rays into the sample. Filters can also be used to suppress unwanted spectral components. For example, K_β radiation is often reduced by filtering because it can create additional weak diffraction peaks. If such peaks are not recognised, they may be incorrectly interpreted as impurity phases or additional crystal orientations.

In this experiment, weak spectral artefacts were considered during the assignment of the ZnO thin-film diffraction patterns. This was particularly important because epitaxial thin-film scans contain only a small number of strong reflections, making automatic database matching less reliable than for polycrystalline samples.

2.4 Bragg Diffraction

X-ray diffraction by a crystal occurs when waves scattered from parallel lattice planes interfere constructively. Constructive interference is described by Bragg's law,

$$n\lambda = 2d \sin\theta, \tag{2}$$

where n is the diffraction order, λ is the X-ray wavelength, d is the lattice-plane spacing, and θ is the Bragg angle.

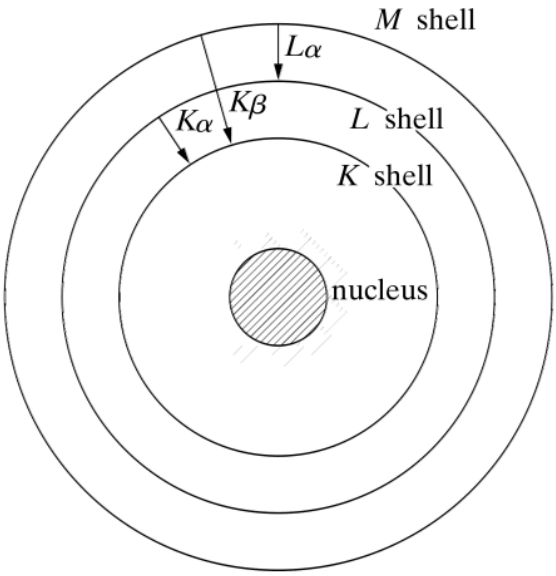


Figure 3: Geometry of Bragg diffraction from parallel lattice planes separated by the distance d. Constructive interference occurs when the path difference between rays scattered from neighbouring planes equals an integer multiple of the wavelength. Ref. [2].

In the measured diffraction patterns, the horizontal axis is usually given as 2θ, which is the angle between the incident and diffracted beam. Therefore, the Bragg angle used in the calculation is

$$\theta = \frac{1}{2}(2\theta). \tag{3}$$

For first-order diffraction, $n = 1$, the lattice spacing can be obtained from the measured peak position by

$$d = \frac{\lambda}{2\sin\theta}. \tag{4}$$

Thus, each diffraction peak gives information about the spacing of a specific family of lattice planes. By comparing the measured peak positions with reference data, the corresponding lattice planes and crystal phases can be identified.

2.5 Crystal Planes and Miller Indices

Crystal planes are described by Miller indices (hkl). These indices define the orientation of a set of parallel lattice

planes within the crystal lattice. Each diffraction peak can be assigned to a particular set of lattice planes if its measured spacing agrees with the expected spacing for that reflection.

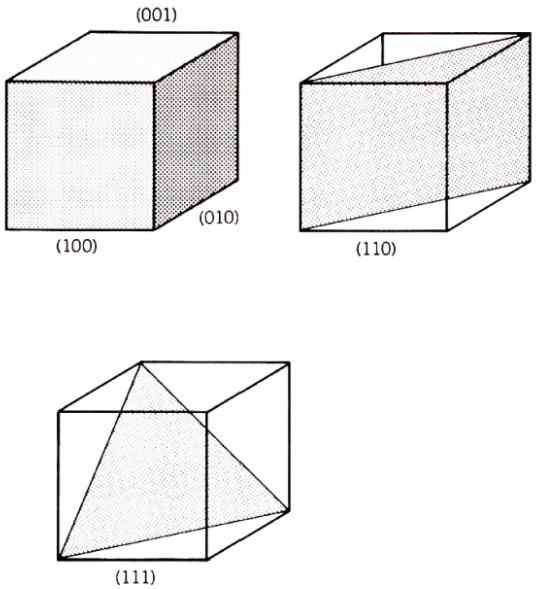


Figure 4: Examples of lattice planes in a simple cubic crystal and their corresponding Miller indices. The figure illustrates how different plane orientations are labelled by different (hkl) values. Ref. [2].

For hexagonal crystals, a four-index notation is often used. This is called Miller–Bravais notation and is written as $(hkil)$. The additional index i is not independent, but is related to the first two indices by

$$i = -(h + k). \tag{5}$$

This notation is useful for hexagonal materials such as ZnO and sapphire because it makes the symmetry of the basal plane more explicit. In this report, ZnO reflections were sometimes discussed using both the three-index notation from the database and the four-index notation commonly used for hexagonal crystals. For example, the ZnO (002) reflection corresponds to the (0002) basal-plane reflection, while the ZnO (110) reflection corresponds to the $(11\bar{2}0)$ reflection in four-index notation.

2.6 Lattice Spacing and Lattice Constants

The relation between lattice spacing and lattice constants depends on the crystal structure. For cubic crystals, the spacing of the (hkl) lattice planes is given by

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}, \tag{6}$$

where a is the cubic lattice constant. Therefore, the lattice constant can be calculated from a measured lattice spacing by

$$a = d_{hkl}\sqrt{h^2 + k^2 + l^2}. \tag{7}$$

For hexagonal crystals, two lattice constants are required: the basal-plane lattice constant a and the out-of-plane lattice constant c. The lattice-spacing relation is

$$\frac{1}{d_{hkl}^2} = \frac{4}{3}\frac{h^2 + hk + k^2}{a^2} + \frac{l^2}{c^2}. \tag{8}$$

This relation was especially important for ZnO, which crystallises in the hexagonal wurtzite structure. Reflections with $l = 0$ mainly provide information about the lattice constant a, while reflections of the type $(000l)$ directly probe the c lattice constant. For a $(000l)$ reflection, Eq. (8) simplifies to

$$d_{000l} = \frac{c}{l}, \tag{9}$$

so that

$$c = l d_{000l}. \tag{10}$$

Equation (10) was used to determine the out-of-plane lattice parameter of the oriented ZnO thin films.

2.7 Goniometer Errors and Lattice-Constant Extrapolation

The lattice constants calculated from individual diffraction peaks are affected by systematic and statistical errors. Possible sources of systematic angular errors include sample displacement, zero-offset errors of the goniometer, sample transparency, imperfect sample alignment, and instrumental broadening. These effects can shift the measured diffraction angles and therefore lead to slightly different lattice constants for different reflections.

A common way to reduce such systematic errors is to calculate the lattice constant from several reflections and extrapolate the result to

$$\theta = 90^\circ. \tag{11}$$

At high diffraction angles, many angular error contributions become smaller, so the extrapolated value gives a more reliable estimate of the true lattice constant.

In this report, three correction functions were used for the extrapolation. The first was

$$f_1(\theta) = \cos^2\theta. \tag{12}$$

The second was

$$f_2(\theta) = \cot\theta. \tag{13}$$

The third was the Nelson–Riley function,

$$f_{\mathrm{NR}}(\theta) = \frac{1}{2}\left(\frac{\cos^2\theta}{\theta} + \frac{\cos^2\theta}{\sin\theta}\right). \tag{14}$$

For each correction function, the calculated lattice constants were plotted against $f(\theta)$ and fitted linearly. The corrected lattice constant was then obtained from the intercept at

$$f(\theta) = 0, \tag{15}$$

which corresponds to $\theta = 90°$.

2.8 Polycrystalline Samples and Epitaxial Thin Films

A polycrystalline sample consists of many small crystallites with different orientations. In a symmetric 2θ–ω scan, many different lattice-plane families can satisfy Bragg's condition because the crystallites are randomly oriented. Therefore, polycrystalline samples usually show several allowed reflections in the diffraction pattern. This makes them suitable for phase identification and lattice-constant determination.

Epitaxial or strongly textured thin films behave differently. In a symmetric 2θ–ω scan, the observed film reflections mainly correspond to lattice planes that are parallel to the sample surface. As a result, only selected reflections appear strongly in the diffraction pattern. The visible reflection family can therefore be used to determine the out-of-plane crystallographic orientation of the film.

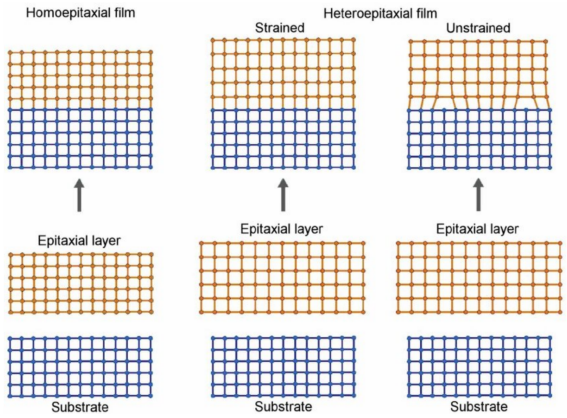


Figure 5: Schematic comparison of epitaxial film growth on a substrate. In heteroepitaxy, lattice mismatch between film and substrate can lead to elastic strain in thin films and to partial strain relaxation through defects or dislocations at larger thickness. Ref. [3].

This distinction was central for the ZnO thin-film analysis. The polycrystalline ZnO reference sample showed many ZnO reflections, while the ZnO thin films on sapphire showed only selected ZnO reflections. The observed thin-film reflections were therefore used to infer the crystallographic growth direction of the films.

2.9 Peak Width, Coherence Length, and Mosaicity

The width of a diffraction peak contains information about the structural quality of the sample. The full width at half maximum, abbreviated as FWHM, is the angular width of a peak at half of its maximum intensity. For a Gaussian peak profile,

$$I(x) = I_0 + A \exp\left[-\frac{(x - x_0)^2}{2\sigma^2}\right], \tag{16}$$

the FWHM is related to the standard deviation σ by

$$\mathrm{FWHM} = 2\sqrt{2 \ln 2}\,\sigma. \tag{17}$$

In a 2θ–ω scan, peak broadening can be caused by variations in lattice-plane spacing, strain inhomogeneity, finite coherent domain size, and instrumental resolution. Therefore, the FWHM of a 2θ–ω peak is related mainly to the distribution of lattice spacings and the coherence of the diffracting crystal regions.

In an ω scan, also called a rocking curve, the detector is kept fixed at the selected 2θ position while the sample angle ω is varied. The FWHM of the rocking curve describes the angular spread of the diffracting lattice planes. A narrow rocking curve indicates good alignment of the crystal planes, while a broad rocking curve indicates stronger tilt mosaicity.

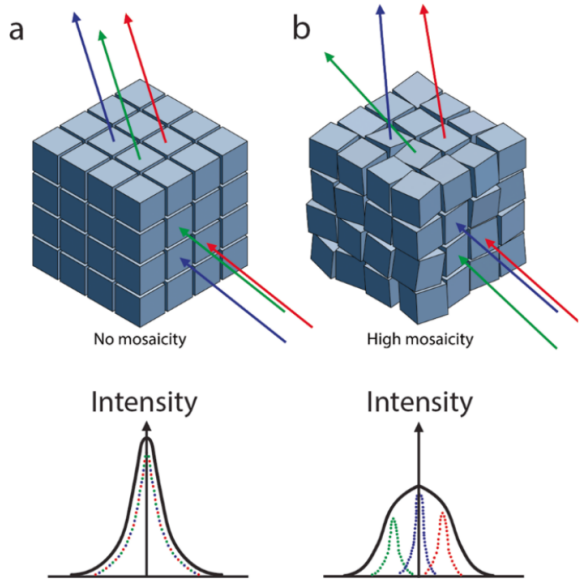


Figure 6: Schematic illustration of tilt mosaicity and its influence on the rocking curve. A well-aligned crystal gives a narrow ω peak, while a larger spread of crystallite orientations produces a broader rocking curve. Ref. [4].

For epitaxial thin films, the comparison between film and substrate rocking curves is especially useful. A high-quality single-crystal substrate is expected to show a narrower ω peak than the thin film. Broader ZnO rocking curves indicate that the film contains a larger spread of crystallite orientations or mosaic blocks relative to the substrate.

3. Experimental Setup and Methods

3.1 X-ray Diffractometer

All diffraction measurements were carried out using a Philips X'Pert X-ray diffractometer. The instrument was equipped with a copper X-ray tube and was operated in Bragg–Brentano geometry. In this geometry, the incident beam, sample surface, and detector were arranged such that the sample angle ω and the detector angle 2θ were coupled during standard diffraction scans. The diffracted intensity was recorded as a function of the diffraction angle.

For the standard diffraction measurements, coupled 2θ–ω scans were performed. During these scans, the detector was moved through the angle 2θ, while the sample was rotated by the corresponding angle $\omega = \theta$. This measurement geometry was used to record diffraction patterns of the polycrystalline samples, the ZnO thin films on sapphire, and the unknown oxide powder.

For selected reflections of the ZnO thin films and sapphire substrates, additional ω scans were performed. In these scans, the detector position was fixed at the chosen 2θ peak position, while only the sample angle ω was varied. These measurements were used to determine rocking-curve widths and to compare the angular alignment of the ZnO films with that of the sapphire substrates.

3.2 Measurement Programme

The experiment consisted of three main types of measurements. First, polycrystalline bulk samples were investigated in order to assign diffraction peaks, determine lattice constants, and apply extrapolation methods for systematic-error correction. In this part, polycrystalline ZnO and copper samples were measured and compared with PC-PDF reference data.

Second, epitaxial ZnO thin films grown on differently oriented sapphire substrates were investigated. One ZnO film was grown on a-plane sapphire, while the other was grown on r-plane sapphire. The 2θ–ω diffraction patterns were used to identify ZnO and sapphire reflections and to determine the out-of-plane crystallographic orientation of the ZnO films. Selected high-angle ZnO and sapphire peaks were then measured by ω scans in order to compare the rocking-curve widths of film and substrate reflections.

Finally, an unknown oxide powder was measured. Its diffraction pattern was compared with candidate reference patterns from the PC-PDF database. The phase identification was based on the agreement between measured and reference peak positions, supported by the relative intensity distribution and crystallographic consistency of the proposed phases.

3.3 Scan Parameters

The angular scan ranges were chosen according to the relevant reflections expected for each sample. The measured data were exported as `.xrdml` files and analysed using the angular axes stored in the measurement files. The main scan parameters are summarised in Table 1.

Table 1: Summary of the measurement parameters used for the X-ray diffraction scans.

Measurement	Scan range	Angular increment	Time per step
Polycrystalline ZnO	30°–130° in 2θ	from exported scan axis	0.5 s
Polycrystalline Cu	40°–140° in 2θ	from exported scan axis	0.5 s
ZnO on a-plane sapphire	30°–130° in 2θ	from exported scan axis	0.5 s
ZnO on r-plane sapphire	20°–130° in 2θ	from exported scan axis	0.5 s
Selected ω scans	approximately 5° in ω	0.01°	0.5 s
Unknown oxide powder	10°–90° in 2θ	from exported scan axis	0.5 s

For the ZnO thin-film rocking-curve measurements, one ZnO reflection and one sapphire reflection were selected for each substrate orientation. The detector was kept fixed at the selected 2θ position, while the sample angle ω was scanned over a range of approximately 5°. The selected reflections and alignment offsets are listed in Table 2.

Table 2: Selected reflections and alignment settings for the ω scans of the ZnO thin films and sapphire substrates.

Sample	Selected reflection	Fixed 2θ / degree	Offset / degree	ω step / degree
ZnO on a-plane sapphire	ZnO (0006)	125.30	0.230	0.01
ZnO on a-plane sapphire	Al_2O_3 $(22\bar{4}0)$	80.72	0.230	0.01
ZnO on r-plane sapphire	ZnO $(11\bar{2}0)$	56.69	0.030	0.01
ZnO on r-plane sapphire	Al_2O_3 $(04\bar{4}8)$	124.69	0.030	0.01

The offsets were used during the alignment of the thin-film samples in the diffractometer software. They accounted for the angular setting required to centre the chosen reflection before the corresponding rocking curve was recorded.

3.4 Peak Assignment and Data Evaluation

The measured diffraction patterns were analysed using X'Pert Data Viewer and X'Pert HighScore. Peak positions were extracted from the measured intensity distributions and compared with reference data from the PC-PDF database. For the polycrystalline samples, the observed reflections were assigned to the corresponding crystal phases and Miller indices. The assigned peak positions were then used to calculate lattice spacings using Eq. (4).

For the cubic copper sample, the lattice constant was calculated using Eq. (7). For the hexagonal ZnO sample, the relation in Eq. (8) was used. Reflections of the type $(000l)$ were used to determine the c lattice constant through Eq. (10). The lattice constants obtained from individual reflections were extrapolated using the correction functions given in Eqs. (12), (13), and (14).

For the ZnO thin films, the measured 2θ–ω diffraction patterns were compared with the polycrystalline ZnO reference pattern. Since the thin films were epitaxial or strongly textured, only selected ZnO reflections appeared

strongly in the symmetric diffraction geometry. These reflections were used to determine the out-of-plane growth direction of the films. Sapphire substrate peaks were assigned separately using the corresponding reference data for Al_2O_3.

Weak spectral artefacts were also considered during the thin-film peak assignment. In particular, possible Cu K_β and W L_α contributions were checked so that weak additional peaks were not incorrectly interpreted as extra crystalline phases or additional ZnO orientations.

The ω scans of the selected ZnO and sapphire reflections were fitted using the Gaussian peak model given in Eq. (16). The full width at half maximum was then obtained from Eq. (17). The resulting rocking-curve widths were used to compare the tilt mosaicity of the ZnO films with that of the sapphire substrates.

For the unknown oxide powder, the measured diffraction pattern was compared with the highest-scoring phase proposals from the PC-PDF database. The final phase assignment was based primarily on peak-position agreement, while relative peak intensities and lattice-parameter consistency were used as additional criteria.

3.5 Uncertainty Treatment

The main uncertainties in the diffraction analysis arose from the finite angular step size, peak-position determination, unresolved or partially resolved $K_{\alpha1}/K_{\alpha2}$ splitting, background fluctuations, and systematic instrumental errors. In the lattice-constant analysis, the uncertainty of the measured diffraction angle propagated into the uncertainty of the lattice spacing calculated from Eq. (4). The corresponding uncertainty in the lattice constants was then obtained by propagating this spacing uncertainty through the relevant crystal-structure relation.

For the extrapolated lattice constants, the spread of the individual reflection values and the uncertainty of the linear-fit intercept were used to estimate the uncertainty of the corrected lattice parameter. The comparison of the $\cos^2\theta$, $\cot\theta$, and Nelson–Riley extrapolations also provided a check on the stability of the correction procedure.

For the ω scans, the peak centre and FWHM were obtained from Gaussian fits. The uncertainty of the fitted width parameter was propagated to the FWHM using Eq. (17). The resulting uncertainty described the statistical uncertainty of the fitted peak width, while possible systematic contributions from instrumental broadening and sample alignment were considered qualitatively in the discussion.

For the unknown oxide powder, the uncertainty was dominated not by numerical fitting errors alone, but also by the reliability of the database match. Therefore, the final phase identification was assessed using the consistency of several measured peaks rather than relying only on the highest database score.

4. Results and Discussion

4.1 Polycrystalline Bulk Samples

4.1.1 Measurement procedure

Polycrystalline ZnO and Cu samples were investigated using the Philips X'Pert diffractometer in Bragg–Brentano geometry. The measurements were performed as 2θ–ω scans, in which the detector angle is varied by 2θ while the sample is rotated by $\omega = \theta$. This geometry is well suited for polycrystalline materials because the randomly oriented crystallites provide many lattice-plane orientations that satisfy Bragg's diffraction condition.

Before each measurement, the X-ray generator was operated at 25 kV and 20 mA, and the scan parameters were configured using the X'Pert Data Collector software. The diffraction patterns were recorded in the angular ranges listed in Table 1 with a step size of 0.04° and a counting time of 0.5 s per step.

The measured diffraction peaks were analysed using X'Pert Data Viewer and X'Pert HighScore. Phase identification and Miller-index assignment were carried out by comparison with reference patterns from the PC-PDF database. For ZnO, the hexagonal wurtzite structure (reference card 00-036-1451) was identified, whereas the Cu sample was assigned to the face-centred cubic (fcc) copper phase (reference card 00-004-0836).

Lattice constants were calculated from the measured peak positions using Bragg's law and the corresponding crystal-structure relations. To reduce systematic angular errors, the lattice parameters obtained from individual reflections were extrapolated to $\theta = 90°$ using the $\cos^2\theta$, $\cot\theta$, and Nelson–Riley correction functions.

4.1.2 Results

ZnO

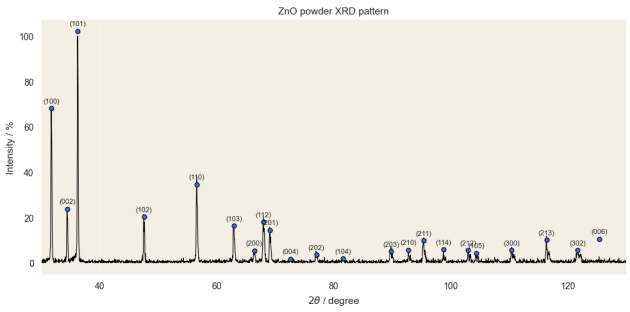


Figure 7: Measured 2θ-ω diffraction pattern of the polycrystalline ZnO sample

Measured diffraction pattern The measured diffraction pattern exhibits numerous sharp Bragg reflections characteristic of a highly crystalline polycrystalline ZnO sample. The strongest reflection occurs at approximately $2\theta = 36.18°$, corresponding to the (101) reflection of hexagonal ZnO.

Comparison with the PC-PDF database shows excellent agreement with reference card 00-036-1451 (hexagonal wurtzite ZnO). All major reflections could be assigned unambiguously.

Table 3: Selected diffraction peaks of the ZnO sample.

(hkl)	2θ / °	θ / °	d / Å	Rel. intensity / %	FWHM / °
(100)	31.68	15.84	2.82	65.1	0.147
(002)	34.38	17.19	2.61	19.1	0.166
(101)	36.18	18.09	2.48	100.0	0.146
(102)	47.54	23.77	1.91	17.8	0.154
(110)	56.53	28.27	1.63	37.3	0.156
(103)	62.88	31.44	1.48	24.0	0.156
(200)	66.38	33.19	1.41	4.1	0.161
(112)	67.96	33.98	1.38	19.7	0.164
(201)	69.10	34.55	1.36	8.6	0.165
(004)	72.56	36.28	1.30	2.0	0.170

Peak assignment The measured peak positions are in excellent agreement with the PC-PDF reference pattern. The strongest reflection corresponds to the (101) peak, which is also the most intense reflection in the reference data.

The complete peak assignment contained additional weak high-angle reflections up to approximately $2\theta = 126°$. For readability, only the most prominent reflections are listed in Table 3. All successfully assigned reflections were nevertheless included in the lattice-constant determination and extrapolation procedure.

Lattice constants For reflections with $l = 0$, the lattice constant a was calculated directly. For reflections with $h = k = 0$, the lattice constant c was determined. The Cu $K_{\alpha 1}$ wavelength $\lambda = 1.5406$ Å was used throughout the calculations.

Table 4: Extrapolated lattice constants of ZnO obtained using different correction functions.

Extrapolation function	a / Å	c / Å
$\cos^2\theta$	3.2471 ± 0.0024	5.2015 ± 0.0011
$\cot\theta$	3.2475 ± 0.0010	5.2021 ± 0.0012
Nelson–Riley	3.2480 ± 0.0008	5.2022 ± 0.0010
PC-PDF value	3.2498	5.2066

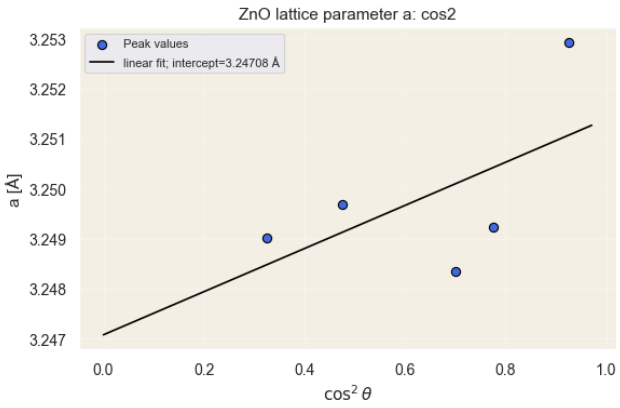


Figure 8: Extrapolation of the ZnO lattice parameter a as a function of $\cos^2\theta$. The linear fit was extrapolated to $\cos^2\theta = 0$, corresponding to $\theta = 90°$, in order to reduce systematic angular errors and obtain the corrected lattice constant.

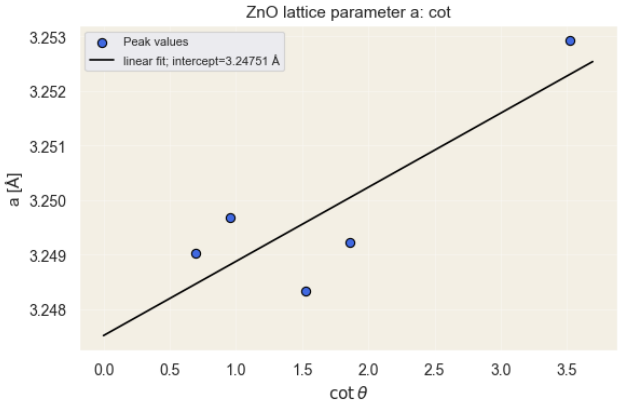


Figure 9: Extrapolation of the ZnO lattice parameter a using the $\cot\theta$ correction function. The intercept of the linear fit at $\cot\theta = 0$ yields the corrected value of the lattice parameter.

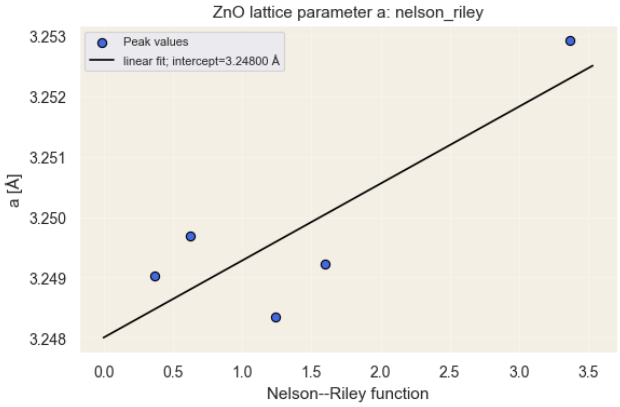


Figure 10: Nelson–Riley extrapolation of the ZnO lattice parameter a. This correction method combines trigonometric functions of θ and is commonly used to minimise systematic goniometer and sample-displacement errors in powder X-ray diffraction measurements.

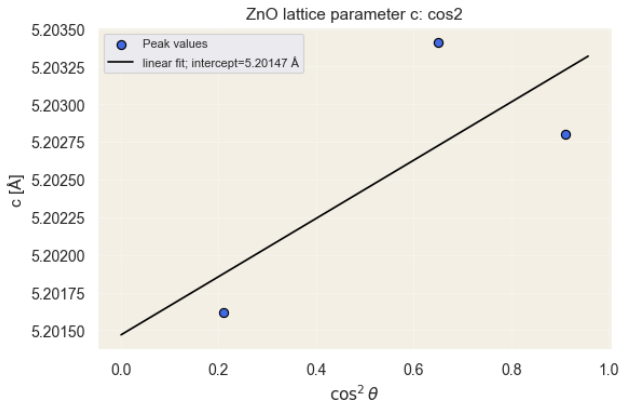


Figure 11: Extrapolation of the ZnO lattice parameter c as a function of $\cos^2\theta$. The extrapolated intercept at $\theta = 90°$ provides the corrected value of the out-of-plane lattice parameter.

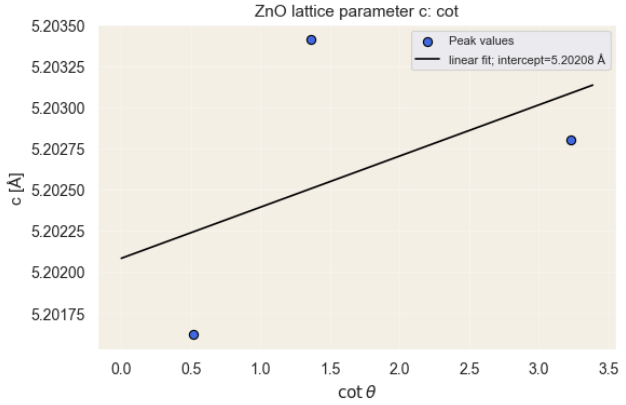


Figure 12: Extrapolation of the ZnO lattice parameter c using the $\cot\theta$ correction function. The resulting intercept corresponds to the corrected lattice parameter after compensation of systematic angular errors.

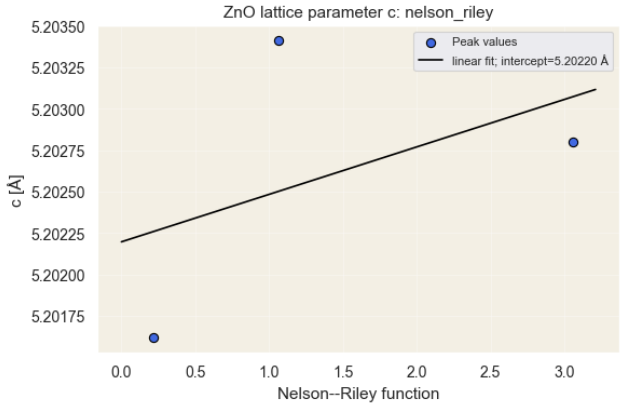


Figure 13: Nelson–Riley extrapolation of the ZnO lattice parameter c. The corrected lattice parameter is obtained from the intercept of the linear regression at a Nelson–Riley value of zero.

All three extrapolation methods yield highly consistent values. The Nelson–Riley extrapolation gives

$$a = (3.2480 \pm 0.0008)\ \text{Å}$$

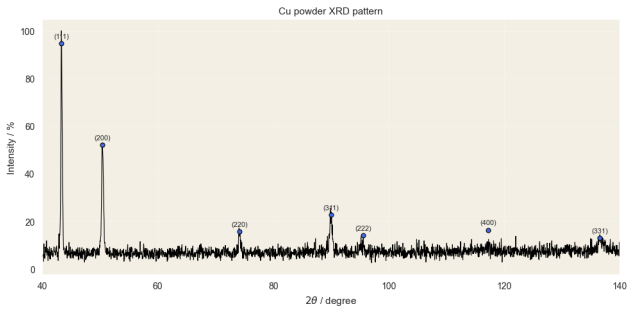


Figure 14: Measured 2θ-ω diffraction pattern of the polycrystalline copper sample

and

$$c = (5.2022 \pm 0.0010)\ \text{Å}.$$

Compared with the PC-PDF values

$$a_{\text{ref}} = 3.2498\ \text{Å}, \qquad c_{\text{ref}} = 5.2066\ \text{Å},$$

the deviations are approximately 0.06% for a and 0.08% for c, indicating very good agreement between experiment and literature.

Copper The measured copper diffraction pattern shows sharp Bragg reflections at positions consistent with an fcc copper phase. The strongest reflection occurs at approximately $2\theta = 43.34°$, corresponding to the (111) reflection.

Table 5: Assigned diffraction peaks of the copper sample.

(hkl)	2θ / °	θ / °	d / Å	Rel. intensity / %	FWHM / °
(111)	43.34	21.67	2.0859	100.0	0.264
(200)	50.44	25.22	1.8078	51.4	0.372
(220)	74.16	37.08	1.2776	10.1	0.396
(311)	89.99	45.00	1.0894	15.9	0.440
(222)	95.59	47.80	1.0399	5.9	0.047
(400)	117.27	58.63	0.9021	9.2	0.024
(331)	136.58	68.29	0.8291	4.1	0.423

For the cubic copper structure, the lattice constant was calculated from each assigned reflection using

$$a = d_{hkl}\sqrt{h^2 + k^2 + l^2}.$$

Table 6: Extrapolated lattice constant of copper obtained using different correction functions.

Extrapolation function	a / Å
$\cos^2\theta$	3.6086 ± 0.0042
$\cot\theta$	3.6087 ± 0.0036
Nelson–Riley	3.6093 ± 0.0029
PC-PDF value	3.6150

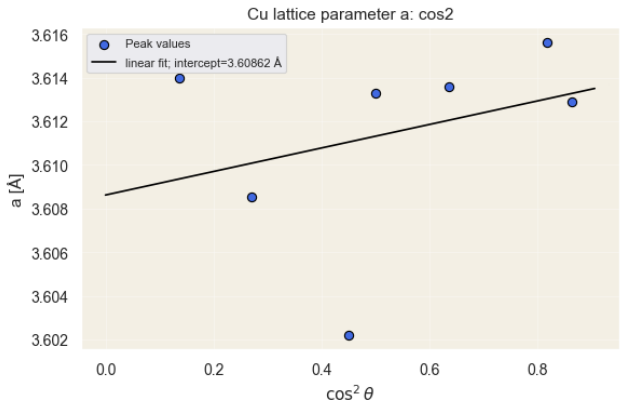


Figure 15: Determination of the copper lattice constant a using the $\cos^2\theta$ extrapolation method. The lattice constants calculated from individual Cu reflections were plotted as a function of $\cos^2\theta$ and fitted linearly. Extrapolation to $\cos^2\theta = 0$ ($\theta = 90°$) yields the corrected lattice constant and reduces systematic angular errors.

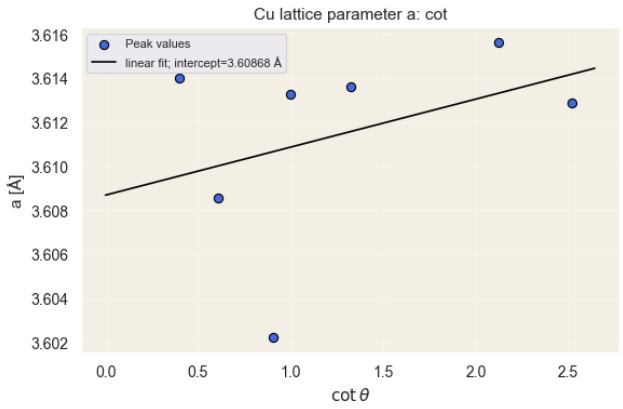


Figure 16: Determination of the copper lattice constant a using the $\cot\theta$ extrapolation method. The intercept of the linear fit at $\cot\theta = 0$ corresponds to the corrected lattice constant after compensation of systematic diffraction-angle errors.

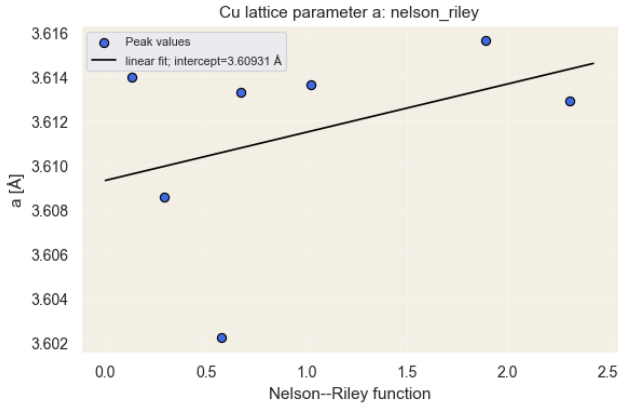


Figure 17: Nelson–Riley extrapolation of the copper lattice constant a. This correction function combines trigonometric terms of the Bragg angle and is commonly used in powder X-ray diffraction to minimise systematic goniometer and sample-displacement errors. The extrapolated intercept provides the final corrected lattice constant of the copper sample.

The extrapolated lattice constants are close to the PC-PDF reference value $a_{\text{ref}} = 3.6150$ Å. The Nelson–Riley extrapolation gives

$$a = (3.6093 \pm 0.0029)\ \text{Å}.$$

The relative deviation from the database value is approximately 0.157%, indicating good agreement between the measured copper diffraction pattern and the reference fcc copper structure.

4.1.3 Discussion

Both measured diffraction patterns exhibit sharp and well-defined Bragg peaks, indicating that the investigated ZnO and Cu samples are highly crystalline. Since the samples are polycrystalline, multiple reflections are observed, corresponding to different families of lattice planes that satisfy Bragg's law.

The measured peak positions show good agreement with the corresponding PC-PDF reference patterns. For ZnO, all major reflections could be assigned to the hexagonal wurtzite structure, with the (101) reflection being the strongest peak. For Cu, the observed reflections are consistent with the expected face-centred cubic structure, dominated by the (111) reflection. No significant additional peaks were detected for either sample, indicating that both materials are predominantly single-phase.

The lattice constants obtained from the individual reflections are mutually consistent and the three extrapolation methods yield very similar results. For ZnO, the extrapolated lattice parameters agree with the literature values within approximately 0.1%, demonstrating excellent agreement between experiment and reference data. For Cu, the extrapolated lattice constant differs from the database value by only about 0.2%, which is also well within the expected accuracy of the experiment.

The remaining deviations can be attributed to systematic effects such as sample displacement, finite instrumental resolution, uncertainties in the fitted peak positions, and residual goniometer alignment errors. In addition, weak high-angle reflections exhibit lower counting statistics and therefore larger uncertainties than the intense low-angle peaks.

Overall, the measured diffraction patterns, peak assignments, and lattice constants are in good agreement with the expected crystal structures of ZnO and Cu. The analysis therefore confirms the successful phase identification and lattice-parameter determination for both polycrystalline samples.

4.2 Epitaxial ZnO Thin Films on Sapphire

4.2.1 Measurement Procedure

Two epitaxial ZnO thin films grown on differently oriented sapphire substrates were investigated. One film was deposited on a-plane sapphire, while the other film was deposited on r-plane sapphire. The measurements were carried out using the Philips X'Pert diffractometer in Bragg–Brentano geometry with Cu radiation. The diffraction patterns were first recorded as coupled 2θ–ω scans, in which the detector angle 2θ and the sample angle ω were moved simultaneously.

For the ZnO film on a-plane sapphire, the diffraction pattern was recorded in the range from 30° to 130° in 2θ. For the ZnO film on r-plane sapphire, the scan range was extended to lower angles and covered 20° to 130° in 2θ. A counting time of 0.5 s per step was used. The exported angular axis from the `.xrdml` files was used for all numerical analysis.

Table 7: Measurement parameters used for the coupled 2θ–ω scans of the ZnO thin films.

Sample	Scan range in 2θ	Scan type	Time per step
ZnO on a-plane sapphire	30°–130°	coupled 2θ–ω	0.5 s
ZnO on r-plane sapphire	20°–130°	coupled 2θ–ω	0.5 s

The resulting diffraction patterns were analysed using X'Pert Data Viewer and X'Pert HighScore. The measured peak positions were compared with PC-PDF reference patterns for ZnO and sapphire. The polycrystalline ZnO pattern measured previously was used as a reference for randomly oriented ZnO. This comparison allowed the thin-film reflections to be distinguished from substrate reflections and made it possible to determine the out-of-plane orientation of the ZnO films. Weak spectral contributions such as Cu K_β and W L_α lines were also considered during the peak assignment so that they were not misinterpreted as additional crystalline phases or extra ZnO orientations.

After the main reflections had been assigned, selected high-angle ZnO and sapphire peaks were measured again using ω scans. In these scans, the detector was kept fixed at the selected 2θ peak position while only the sample angle ω was varied. This allowed the rocking curves of the ZnO film peaks and sapphire substrate peaks to be measured. The ω scans covered an angular range of approximately 5° with a step size of 0.01° and a counting time of 0.5 s per step.

Table 8: Selected reflections and scan settings used for the ω scans of the ZnO films and sapphire substrates.

Sample	Selected reflection	Fixed 2θ / degree	Offset / degree	ω step / degree
ZnO on a-plane sapphire	ZnO (0006)	125.30	0.230	0.01
ZnO on a-plane sapphire	Al_2O_3 ($22\bar{4}0$)	80.72	0.230	0.01
ZnO on r-plane sapphire	ZnO ($11\bar{2}0$)	56.69	0.030	0.01
ZnO on r-plane sapphire	Al_2O_3 ($04\bar{4}8$)	124.69	0.030	0.01

The offsets listed in Table 8 were applied during alignment in the diffractometer software in order to centre the selected reflections before recording the rocking curves. The measured ω scans were later fitted with the Gaussian peak model from Eq. (16), and the corresponding FWHM values were calculated using Eq. (17). The resulting widths were used to compare the crystalline alignment of the ZnO films with that of the sapphire substrates.

4.2.2 Results

The measured 2θ–ω diffraction patterns of the ZnO thin films on a-plane and r-plane sapphire showed clear differences from the polycrystalline ZnO reference pattern. In the polycrystalline sample, many ZnO reflections were visible because many crystallite orientations contributed to the diffraction pattern. In the thin-film samples, however, only selected ZnO reflections appeared strongly. This indicated that the ZnO layers were not randomly oriented, but showed preferred crystallographic orientations with respect to the sapphire substrates.

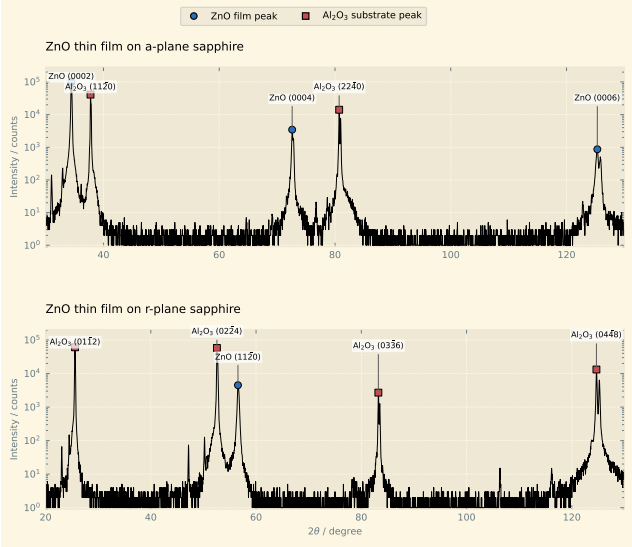


Figure 18: Measured 2θ–ω diffraction patterns of the ZnO thin films on a-plane and r-plane sapphire. The relevant ZnO film reflections and sapphire substrate reflections are marked. Weak Cu K_β and W L_α artefacts were also identified where necessary.

For the ZnO film grown on a-plane sapphire, the main ZnO reflections were observed at approximately 34.44°, 72.60°, and 125.32°. These peaks were assigned to the ZnO (0002), (0004), and (0006) reflections, respectively. The sapphire substrate reflections were observed at approximately 37.76° and 80.72°. Additional weak peaks near the ZnO (0002) region were assigned to Cu K_β and W L_α spectral artefacts rather than to additional ZnO orientations or impurity phases.

Table 9: Peak assignment for the ZnO film on a-plane sapphire.

2θ / degree	Phase	Assigned reflection	Radiation line	Comment
31.04	ZnO	(0002)	Cu K_β	spectral artefact
32.96	ZnO	(0002)	W L_α	spectral artefact
34.44	ZnO	(0002)	Cu $K_{\alpha1}$	real film peak
37.76	Al_2O_3	$(11\bar{2}0)$	Cu $K_{\alpha1}$	substrate peak
72.60	ZnO	(0004)	Cu $K_{\alpha1}$	real film peak
72.80	ZnO	(0004)	Cu $K_{\alpha2}$	doublet shoulder
80.72	Al_2O_3	$(22\bar{4}0)$	Cu $K_{\alpha1}$	substrate peak
80.96	Al_2O_3	$(22\bar{4}0)$	Cu $K_{\alpha2}$	doublet shoulder
125.32	ZnO	(0006)	Cu $K_{\alpha1}$	real film peak
125.84	ZnO	(0006)	Cu $K_{\alpha2}$	doublet shoulder

For the ZnO film grown on r-plane sapphire, the symmetric 2θ–ω scan showed one clear ZnO film reflection at approximately 56.56°. This reflection was assigned to ZnO $(11\bar{2}0)$, corresponding to the (110) reflection in the three-index notation used in the database reference pattern. The remaining intense reflections were assigned mainly to the sapphire substrate. Weak W L_α contributions were identified near selected sapphire reflections and were not treated as separate crystalline phases.

Table 10: Peak assignment for the ZnO film on r-plane sapphire.

2θ / degree	Phase	Assigned reflection	Radiation line	Comment
24.48	Al_2O_3	$(01\bar{1}2)$	W L_α	spectral artefact
25.60	Al_2O_3	$(01\bar{1}2)$	Cu $K_{\alpha1}$	substrate peak
50.20	Al_2O_3	$(02\bar{2}4)$	W L_α	spectral artefact
52.56	Al_2O_3	$(02\bar{2}4)$	Cu $K_{\alpha1}$	substrate peak
52.72	Al_2O_3	$(02\bar{2}4)$	Cu $K_{\alpha2}$	doublet shoulder
56.56	ZnO	$(11\bar{2}0)$ / (110)	Cu $K_{\alpha1}$	real film peak
83.24	Al_2O_3	$(03\bar{3}6)$	Cu $K_{\alpha1}$	substrate peak
83.52	Al_2O_3	$(03\bar{3}6)$	Cu $K_{\alpha2}$	doublet shoulder
124.68	Al_2O_3	$(04\bar{4}8)$	Cu $K_{\alpha1}$	substrate peak
125.24	Al_2O_3	$(04\bar{4}8)$	Cu $K_{\alpha2}$	doublet shoulder

The film orientation was determined from the ZnO reflections observed in the symmetric 2θ–ω scans. Since this geometry mainly detected lattice planes parallel to the sample surface, the observed ZnO reflection family indicated the out-of-plane crystallographic direction of the film.

Table 11: Crystallographic orientation of the ZnO thin films determined from the observed ZnO reflections.

Sample	Dominant ZnO reflection	Inferred out-of-plane direction
ZnO on a-plane sapphire	(0002), (0004), (0006)	[0001]
ZnO on r-plane sapphire	$(11\bar{2}0)$	$[11\bar{2}0]$

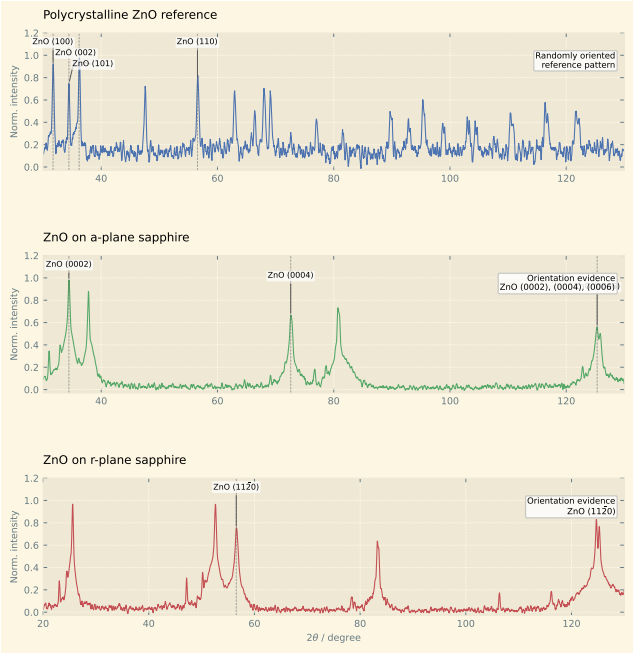


Figure 19: Comparison of the polycrystalline ZnO reference pattern with the ZnO thin-film diffraction patterns. The polycrystalline ZnO sample showed many reflections, while the thin films showed only selected reflections associated with their preferred out-of-plane orientations.

The comparison with the polycrystalline ZnO reference confirmed the preferred orientation of the films. The ZnO film on a-plane sapphire showed only the $(000l)$ reflection family, so it was assigned as c-axis oriented with the [0001] direction normal to the sample surface. The ZnO film on r-plane sapphire showed the $(11\bar{2}0)$ reflection as the dominant ZnO film peak, indicating an out-of-plane direction of $[11\bar{2}0]$.

After the main film and substrate reflections had been assigned, selected high-angle peaks were fitted locally in the 2θ–ω scans and compared with the corresponding ω scans. The local fitting windows were chosen individually for each selected peak. This was necessary because the peaks had different widths, different background levels, and different degrees of overlap with neighbouring Cu $K_{\alpha2}$ shoulders. A single wide fitting range would have included additional spectral contributions and background variations, so the extracted FWHM would no longer have represented only the selected Cu $K_{\alpha1}$ peak. Therefore, each fit was restricted to the local angular region around the corresponding Cu $K_{\alpha1}$ maximum.

Table 12: Comparison of selected peak positions and FWHM values from local 2θ–ω fits and ω-scan fits. The R^2 values refer to the local Gaussian fits of the selected 2θ–ω peaks.

Sample / phase	Reflection	$2\theta_{\mathrm{Cu}\,K_{\alpha1}}$ / degree	$\mathrm{FWHM}_{2\theta-\omega}$ / degree	FWHM_{ω} / degree	R^2
a-plane ZnO	(0006)	125.291 ± 0.003	0.324 ± 0.011	0.40377 ± 0.00109	0.99182
a-plane Al_2O_3	$(22\bar{4}0)$	80.722 ± 0.001	0.0966 ± 0.0026	0.26836 ± 0.00301	0.99797
r-plane ZnO	$(11\bar{2}0)$	56.607 ± 0.003	0.302 ± 0.009	0.75874 ± 0.00178	0.99103
r-plane Al_2O_3	$(04\bar{4}8)$	124.697 ± 0.002	0.160 ± 0.006	0.27571 ± 0.00108	0.98910

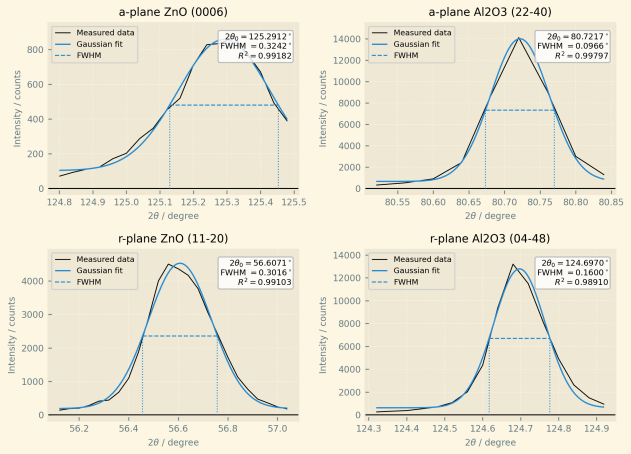


Figure 20: Local Gaussian fits of the selected 2θ–ω peaks. The fits were restricted to angular windows around the Cu $K_{\alpha 1}$ maxima in order to avoid neighbouring Cu $K_{\alpha 2}$ shoulders and unrelated background contributions.

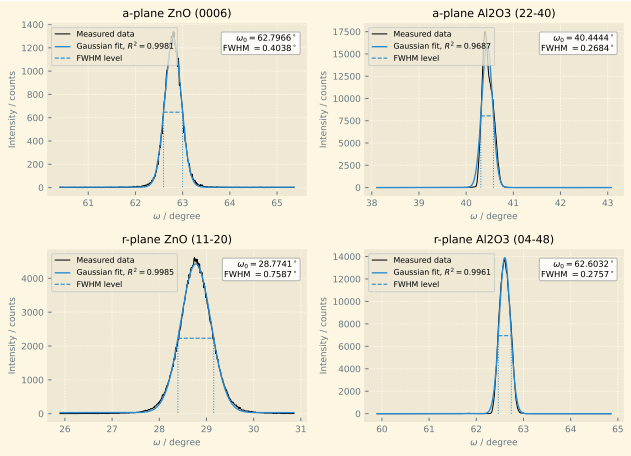


Figure 21: Gaussian fits of the selected ω scans for ZnO film and sapphire substrate reflections. The fitted FWHM values were used to compare the rocking-curve widths of the thin films and substrates.

The local 2θ–ω fits showed that the ZnO film peaks were broader than the corresponding sapphire substrate peaks. For the a-plane sample, the ZnO (0006) reflection had a 2θ–ω FWHM of 0.324°, while the sapphire $(22\bar{4}0)$ reflection had a narrower FWHM of 0.0966°. For the r-plane sample, the ZnO $(11\bar{2}0)$ reflection had a FWHM of 0.302°, while the sapphire $(04\bar{4}8)$ reflection had a FWHM of 0.160°.

The ω-scan fits showed the same general trend. The sapphire rocking curves were narrower than the ZnO rocking curves. The a-plane sapphire peak had an ω-scan FWHM of 0.26836°, while the a-plane ZnO peak had a larger FWHM of 0.40377°. For the r-plane sample, the sapphire peak had an FWHM of 0.27571°, whereas the ZnO peak was much broader, with an FWHM of 0.75874°. This indicated that the ZnO films had a larger tilt spread than the sapphire substrates, with the r-plane ZnO film showing the strongest mosaic broadening.

The out-of-plane lattice parameters of the ZnO films were calculated from the fitted Cu $K_{\alpha 1}$ peak positions using Eq. (4). For the a-plane sample, the ZnO (0006) reflection was used to calculate the c lattice parameter using Eq. (10). For the r-plane sample, the ZnO $(11\bar{2}0)$ reflection was used to determine the hexagonal a lattice parameter, since this reflection corresponds to $a = 2d_{11\bar{2}0}$.

Table 13: Lattice parameters obtained from selected ZnO thin-film reflections and comparison with the polycrystalline ZnO result and PC-PDF reference values.

Sample	ZnO reflection	$2\theta_{\mathrm{Cu}\,K_{\alpha 1}}$ / degree	d / Å	Lattice parameter / Å
ZnO/a-plane sapphire	(0006)	125.291 ± 0.003	0.867277 ± 0.000012	$c = 5.2037 \pm 0.0001$
ZnO/r-plane sapphire	$(11\bar{2}0)$	56.607 ± 0.003	1.624614 ± 0.000074	$a = 3.2492 \pm 0.0002$
Polycrystalline ZnO	extrapolated value	–	–	$a = 3.2480 \pm 0.0008$, $c = 5.2022 \pm 0.0010$
PC-PDF ZnO reference	database value	–	–	$a = 3.2498$, $c = 5.2066$

The ZnO film grown on a-plane sapphire gave an out-of-plane lattice parameter of $c = (5.2037 \pm 0.0001)$ from the (0006) reflection. The ZnO film grown on r-plane sapphire gave $a = (3.2492 \pm 0.0002)$ from the $(11\bar{2}0)$ reflection. Both values were close to the corresponding polycrystalline ZnO and PC-PDF reference values.

4.3 Unknown Oxide Powder

4.3.1 Measurement Procedure

An unknown oxide powder sample (bottle number 4) was prepared and measured using the Philips X'Pert diffractometer in Bragg–Brentano geometry. The diffraction pattern was recorded as a coupled 2θ–ω scan in the range

$$10^\circ \le 2\theta \le 90^\circ,$$

using a step size of 0.04° and a counting time of 0.5 s per step.

Table 14: Measurement parameters for the unknown oxide powder sample.

Quantity	Value
Bottle number	4
Scan type	2θ–ω scan
2θ range	10° to 90°
Step size	0.04°
Time per step	0.5 s

The phase analysis was performed using X'Pert HighScore together with the implemented PC-PDF database. The automatic search routine produced several candidate phases ranked by decreasing score value. The three highest-scoring matches were subsequently compared with respect to peak-position agreement, peak intensities, crystal structure, and lattice parameters.

4.3.2 Results

Measured diffraction pattern The measured diffraction pattern contains several intense reflections in the region between approximately 27° and 57° together with a number

of weaker peaks at higher diffraction angles. The strongest reflection is located near $2\theta \approx 28°$.

Table 15: Three highest-scoring phase proposals obtained from the PC-PDF database search.

Rank	Score	Phase	Crystal system	Reference
1	87	Lanthanum Molybdenum Oxide	Hexagonal	00-034-1041
2	87	Praseodymium Oxide ($Pr_{10}O_{18}$)	Monoclinic	01-083-1194
3	86	Praseodymium Oxide ($Pr_{24}O_{44}$)	Triclinic	01-080-1283

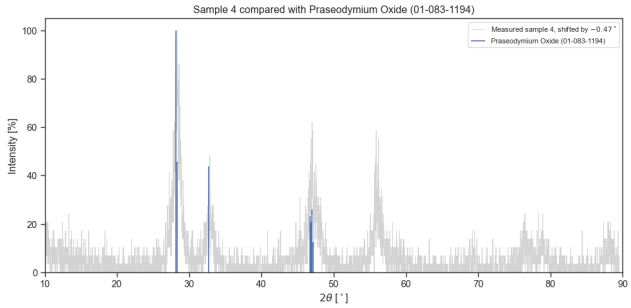


Figure 22: Comparison of the measured diffraction pattern with the Praseodymium Oxide reference pattern (01-083-1194).

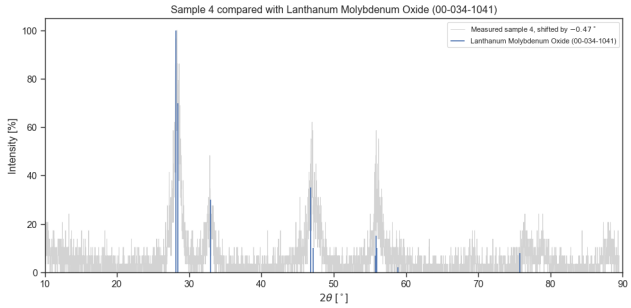


Figure 23: Comparison of the measured diffraction pattern with the Lanthanum Molybdenum Oxide reference pattern (00-034-1041).

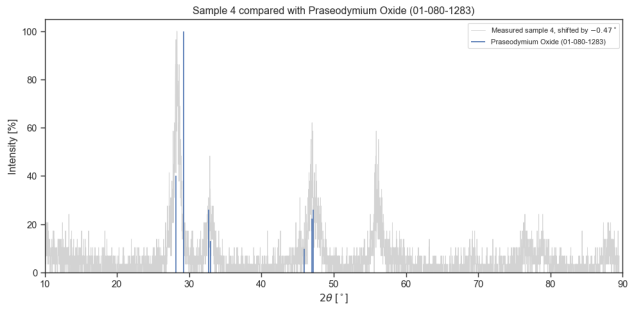


Figure 24: Comparison of the measured diffraction pattern with the Praseodymium Oxide reference pattern (01-080-1283).

Database search results

Peak-position comparison To compare the measured diffraction pattern with the reference patterns, a small systematic angular correction of approximately

$$\Delta(2\theta) \approx -0.47°$$

was applied to the measured data. This correction was estimated from the average shift between the strongest measured reflections and the corresponding reference peaks.

After applying the correction, the strongest reflections of the measured pattern coincide closely with the reference peaks of Lanthanum Molybdenum Oxide.

Table 16: Final phase identification of bottle 4.

Quantity	Result
Bottle number	4
Identified phase	Lanthanum Molybdenum Oxide
Chemical formula	$La_6Mo_2O_{14}$
Reference card	00-034-1041
Crystal system	Hexagonal
Lattice constant a	7.73 Å
Lattice constant c	18.81 Å

4.3.3 Discussion

The phase identification was primarily based on the agreement between the measured and reference peak positions. While all three database matches reproduced the strongest reflection near $2\theta \approx 28°$, differences became apparent when comparing the complete diffraction pattern.

The two Praseodymium Oxide candidates showed reasonable agreement for the strongest peaks but failed to reproduce several secondary reflections. In contrast, Lanthanum Molybdenum Oxide (PC-PDF card 00-034-1041) provided the best overall match. After applying a systematic angular correction of approximately $-0.47°$, the major measured peaks aligned closely with the reference pattern and showed good agreement in relative intensity.

The observed offset is likely caused by instrumental effects such as sample displacement, goniometer zero-shift, or slight alignment errors. Since the shift affects all reflections similarly, it is consistent with a systematic measurement error rather than an incorrect phase assignment.

Most prominent reflections could be assigned to a single phase, and no strong evidence for additional crystalline phases was found. Therefore, bottle number 4 is identified as Lanthanum Molybdenum Oxide ($La_6Mo_2O_{14}$) with hexagonal crystal structure.

5. Conclusion

X-ray diffraction was used to investigate polycrystalline bulk samples, epitaxial ZnO thin films on sapphire, and an unknown oxide powder. The measured diffraction patterns allowed phase identification, lattice-constant determination,

thin-film orientation analysis, and comparison of crystalline quality.

For the polycrystalline samples, the measured peak positions agreed well with the PC-PDF reference patterns. The ZnO sample was identified as hexagonal wurtzite ZnO, while the copper sample was identified as face-centred cubic copper. The extrapolated lattice constants were

$$a_{\mathrm{ZnO}} = (3.2480 \pm 0.0008)\ \text{Å}, \qquad c_{\mathrm{ZnO}} = (5.2022 \pm 0.0010)\ \text{Å},$$

and

$$a_{\mathrm{Cu}} = (3.6093 \pm 0.0029)\ \text{Å}.$$

These values were close to the corresponding PC-PDF reference values, with deviations below about 0.2%. The remaining differences were attributed mainly to systematic effects such as sample displacement, finite instrumental resolution, peak-position uncertainty, and residual goniometer alignment errors.

For the ZnO thin films, the coupled 2θ–ω scans showed that the films were strongly oriented rather than randomly polycrystalline. The ZnO film on a-plane sapphire showed the (0002), (0004), and (0006) reflections and was therefore assigned a [0001] out-of-plane orientation. The ZnO film on r-plane sapphire showed the $(11\bar{2}0)$ reflection as the dominant ZnO film peak and was assigned a $[11\bar{2}0]$ out-of-plane orientation. Local Gaussian fits and ω-scan rocking curves showed that the ZnO film peaks were broader than the sapphire substrate peaks, indicating larger mosaic spread in the films. The r-plane ZnO film showed the strongest tilt broadening, with an ω-scan FWHM of 0.75874°. The obtained ZnO film lattice parameters were

$$c = (5.2037 \pm 0.0001)\ \text{Å}$$

for the a-plane sample and

$$a = (3.2492 \pm 0.0002)\ \text{Å}$$

for the r-plane sample, both close to the reference values.

For the unknown oxide powder, the measured diffraction pattern was compared with PC-PDF database candidates. An angular correction of approximately $-0.47°$ was applied. After this correction, the major measured peaks aligned closely with the Lanthanum Molybdenum Oxide reference pattern and showed good agreement in relative intensity. The observed offset was likely caused by instrumental effects such as sample displacement, goniometer zero-shift, or slight alignment errors. Since the shift affected all reflections similarly, it was consistent with a systematic measurement error rather than an incorrect phase assignment. The unknown sample was therefore identified as hexagonal Lanthanum Molybdenum Oxide, $La_6Mo_2O_{14}$.

Overall, the results showed that X-ray diffraction provided reliable phase identification, lattice-parameter determination, thin-film orientation analysis, and qualitative assessment of crystalline quality for the investigated samples.

References

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Appendix

This appendix contains supplementary diffraction patterns and supporting tables used during the analysis. The main text contains the reduced peak assignments, fitted results, and final interpretations, while the appendix provides additional figures and numerical data for completeness.

A.1 Peak Tables

Table A1: Selected assigned diffraction peaks of the polycrystalline ZnO sample.

(hkl)	2θ / degree	θ / degree	d / Å	Relative intensity / %	FWHM / degree
(100)	31.68	15.84	2.82	65.1	0.147
(002)	34.38	17.19	2.61	19.1	0.166
(101)	36.18	18.09	2.48	100.0	0.146
(102)	47.54	23.77	1.91	17.8	0.154
(110)	56.53	28.27	1.63	37.3	0.156
(103)	62.88	31.44	1.48	24.0	0.156
(200)	66.38	33.19	1.41	4.1	0.161
(112)	67.96	33.98	1.38	19.7	0.164
(201)	69.10	34.55	1.36	8.6	0.165
(004)	72.56	36.28	1.30	2.0	0.170

Table A2: Assigned diffraction peaks of the polycrystalline copper sample.

(hkl)	2θ / degree	θ / degree	d / Å	Relative intensity / %	FWHM / degree
(111)	43.34	21.67	2.0859	100.0	0.264
(200)	50.44	25.22	1.8078	51.4	0.372
(220)	74.16	37.08	1.2776	10.1	0.396
(311)	89.99	45.00	1.0894	15.9	0.440
(222)	95.59	47.80	1.0399	5.9	0.047
(400)	117.27	58.63	0.9021	9.2	0.024
(331)	136.58	68.29	0.8291	4.1	0.423

Table A3: Assigned peaks of the ZnO thin film on a-plane sapphire.

2θ / degree	Phase	Assigned reflection	Radiation line	Comment
31.04	ZnO	(0002)	Cu K_β	spectral artefact
32.96	ZnO	(0002)	W L_α	spectral artefact
34.44	ZnO	(0002)	Cu $K_{\alpha1}$	film peak
37.76	Al_2O_3	$(11\bar{2}0)$	Cu $K_{\alpha1}$	substrate peak
72.60	ZnO	(0004)	Cu $K_{\alpha1}$	film peak
72.80	ZnO	(0004)	Cu $K_{\alpha2}$	doublet shoulder
80.72	Al_2O_3	$(22\bar{4}0)$	Cu $K_{\alpha1}$	substrate peak
80.96	Al_2O_3	$(22\bar{4}0)$	Cu $K_{\alpha2}$	doublet shoulder
125.32	ZnO	(0006)	Cu $K_{\alpha1}$	film peak
125.84	ZnO	(0006)	Cu $K_{\alpha2}$	doublet shoulder

Table A4: Assigned peaks of the ZnO thin film on r-plane sapphire.

2θ / degree	Phase	Assigned reflection	Radiation line	Comment
24.48	Al_2O_3	$(01\bar{1}2)$	W L_α	spectral artefact
25.60	Al_2O_3	$(01\bar{1}2)$	Cu $K_{\alpha1}$	substrate peak
50.20	Al_2O_3	$(02\bar{2}4)$	W L_α	spectral artefact
52.56	Al_2O_3	$(02\bar{2}4)$	Cu $K_{\alpha1}$	substrate peak
52.72	Al_2O_3	$(02\bar{2}4)$	Cu $K_{\alpha2}$	doublet shoulder
56.56	ZnO	$(11\bar{2}0)$ / (110)	Cu $K_{\alpha1}$	film peak
83.24	Al_2O_3	$(03\bar{3}6)$	Cu $K_{\alpha1}$	substrate peak
83.52	Al_2O_3	$(03\bar{3}6)$	Cu $K_{\alpha2}$	doublet shoulder
124.68	Al_2O_3	$(04\bar{4}8)$	Cu $K_{\alpha1}$	substrate peak
125.24	Al_2O_3	$(04\bar{4}8)$	Cu $K_{\alpha2}$	doublet shoulder

A.2 Lattice-Constant and FWHM Summary Tables

Table A5: Extrapolated lattice constants of polycrystalline ZnO obtained using different correction functions.

Extrapolation function	a / Å	c / Å
$\cos^2\theta$	3.2471 ± 0.0024	5.2015 ± 0.0011
$\cot\theta$	3.2475 ± 0.0010	5.2021 ± 0.0012
Nelson–Riley	3.2480 ± 0.0008	5.2022 ± 0.0010
PC-PDF value	3.2498	5.2066

Table A6: Extrapolated lattice constant of polycrystalline copper obtained using different correction functions.

Extrapolation function	a / Å
$\cos^2\theta$	3.6086 ± 0.0042
$\cot\theta$	3.6087 ± 0.0036
Nelson–Riley	3.6093 ± 0.0029
PC-PDF value	3.6150

Table A7: Comparison of selected peak positions and FWHM values from local 2θ–ω fits and ω-scan fits.

Sample / phase	Reflection	$2\theta_{\mathrm{Cu}\,K_{\alpha1}}$ / degree	$\mathrm{FWHM}_{2\theta-\omega}$ / degree	FWHM_{ω} / degree	R^2
a-plane ZnO	(0006)	125.291 ± 0.003	0.324 ± 0.011	0.40377 ± 0.00109	0.99182
a-plane Al_2O_3	$(22\bar{4}0)$	80.722 ± 0.001	0.0966 ± 0.0026	0.26836 ± 0.00301	0.99797
r-plane ZnO	$(11\bar{2}0)$	56.607 ± 0.003	0.302 ± 0.009	0.75874 ± 0.00178	0.99103
r-plane Al_2O_3	$(04\bar{4}8)$	124.697 ± 0.002	0.160 ± 0.006	0.27571 ± 0.00108	0.98910

Table A8: Lattice parameters obtained from selected ZnO thin-film reflections.

Sample	ZnO reflection	$2\theta_{\mathrm{Cu}\,K_{\alpha1}}$ / degree	d / Å	Lattice parameter / Å
ZnO/a-plane sapphire	(0006)	125.291 ± 0.003	0.867277 ± 0.000012	$c = 5.2037 \pm 0.0001$
ZnO/r-plane sapphire	$(11\bar{2}0)$	56.607 ± 0.003	1.624614 ± 0.000074	$a = 3.2492 \pm 0.0002$
Polycrystalline ZnO	extrapolated value	–	–	$a = 3.2480 \pm 0.0008$, $c = 5.2022 \pm 0.0010$
PC-PDF ZnO reference	database value	–	–	$a = 3.2498$, $c = 5.2066$

A.3 HighScore Search Results for the Unknown Oxide

Table A9: Three highest-scoring phase proposals obtained from the PC-PDF database search.

Rank	Score	Phase	Crystal system	Reference
1	87	Lanthanum Molybdenum Oxide	Hexagonal	00-034-1041
2	87	Praseodymium Oxide ($Pr_{10}O_{18}$)	Monoclinic	01-083-1194
3	86	Praseodymium Oxide ($Pr_{24}O_{44}$)	Triclinic	01-080-1283

Table A10: Final phase identification of the unknown oxide powder sample.

Quantity	Result
Bottle number	4
Identified phase	Lanthanum Molybdenum Oxide
Chemical formula	$La_6Mo_2O_{14}$
Reference card	00-034-1041
Crystal system	Hexagonal
Lattice constant a	7.73
Lattice constant c	18.81
Applied angular correction	approximately -0.47°

A.4 Reference Database Values

Table A11: Reference values used for comparison with the measured and calculated results.

Material	Crystal system	a / Å	c / Å
Cu	cubic, fcc	3.6150	–
ZnO	hexagonal, wurtzite	3.2498	5.2066
$La_6Mo_2O_{14}$	hexagonal	7.73	18.81

A.5 Use of AI Assistance

ChatGPT was used as an auxiliary tool during the preparation of this report. It was used to support the structure of the report, improve wording and clarity, identify inconsistencies in the draft, and provide guidance for data analysis and graph plotting. All experimental data, numerical calculations, plots, final interpretations, and conclusions were checked and verified by the authors. ChatGPT was not used as a primary scientific source for the physical theory or reference values.